First full Hamiltonian stochastic runs

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Abstract

We couple the stochastic viscous step in the particle representation (`stochrep.tex`) to the Hamiltonian (ideal SPH) step and run a uniform two-dimensional fluid at rest, starting from zero momenta. This note records the setup and, for each test, what we found: the residual move is needed to keep the lattice modes under control; with it, the fluid thermalizes and stays stable, apart from a slow drift of the energies; the drift comes from the stochastic step, not from the integrator, and the gradient-corrected change of representation reduces it.

I. SETUP

a. The time step. Each time step is that of `generic_step!`, with the stochastic step in place of the deterministic viscous step:

1. a stochastic viscous half step, of length $\Delta t/2$;

2. the leapfrog (drift–kick–drift) of the ideal SPH Hamiltonian H, of length Δt;

3. a second stochastic half step.

The stochastic step is `stochrep_step!` of `stochrep.tex`: a stress sweep and a residual sweep on a checkerboard of cells, at fixed positions. The order of the two sweeps alternates from step to step. Before each step the particle energies e_a and volumes $V_a = s_a/S_a$ are recomputed from the smoothed state, the coefficients are evaluated at the coarse-grained velocity $\dot{\boldsymbol{x}}_a$, and after it the entropies are set by the back map.

b. Parameters.

fluid	conformal, $c_s^2 = 1/3$, $\eta/s = 0.2$, $\zeta = 0$, three dimensions ($d = 3$, ζ_{eff})
box	periodic, 0.6×0.589
lattice	hexagonal, $h = 0.05$
$h/a = 2$	$N = 672$ particles, 56 bonds per star
$h/a = 2.5$	$N = 1020$ particles, 88 bonds per star
entropy density	$S = 10^6$: $T = 90.9$, $s_a = 541$ ($h/a = 2$), 346 ($h/a = 2.5$)
thermal velocity	$\langle \delta v^2 \rangle^{1/2} = 0.043$ per component ($h/a = 2$), 0.054 ($h/a = 2.5$)
time step	$\Delta t = 8.66 \times 10^{-3}$, half the default viscous step
initial state	uniform, at rest, all momenta zero

c. Variants of the viscous step.

- *Stress and residual moves,* $\lambda = 1$: detailed balance, every mode fluctuates.
- *Stress move and deterministic residual,* $\lambda = \infty$: the lattice modes are damped without noise and stay cold.
- *Stress move only.*

The vote is either per bond (the bond split of `stochrep.tex`) or a single vote of the star. The change of representation is either the plain exact map, which holds e_a fixed, or the gradient-corrected map of `changerep.tex`, which holds $e_a - g_a$ fixed and refits the gradient energy g_a afterwards.

d. What is measured.

- The kinetic energy, $\sum_a |\boldsymbol{m}_a|^2$, divided by its equilibrium value $2(N-1)V_a wT$.
- The momentum spectrum, $\langle|\boldsymbol{m}_{\boldsymbol{k}}|^2\rangle/(NV_a wT)$, longitudinal and transverse, over the second half of the run.
- The smoothed energy H, conserved by the leapfrog, and the sum of the particle energies $\sum_a e_a$, held fixed within a stochastic step.
- The rms displacement of the particles from their initial sites, after removing the mean, and the *local disorder*: the rms change of the distances between original nearest neighbours. Both are in units of the lattice spacing a. The first counts collective motion; the second does not.
- The acceptance of the moves.

Scripts: `examples/stochrep_hamiltonian.jl` and its plotting scripts.

II. IS THE RESIDUAL MOVE NEEDED?

The stress move alone already satisfies detailed balance, so $e^{\mathbb{S}}$ is stationary without the residual move. The question is whether the residual move is needed in practice.

a. What the stress move sees. With the stress move alone, a mode is reached only through the fitted gradient. It is damped, and receives noise, at the resolved rate

$$\Gamma_{\mathrm{res}}(k) = R(k)\,\Gamma(k), \qquad R(k) = \frac{K_{\mathrm{res}}(k)}{K(k)}, \tag{1}$$

instead of the rate $\Gamma(k)$ of the bond form. Fig. 1: the resolved fraction R depends only on kh, not on the lattice spacing, in 1D and 2D. It is close to one for $kh \ll 1$, above 0.9 for $kh \lesssim 0.6$, equal to one half at $kh \simeq 1.6$, and negligible beyond $kh \simeq 4$. The zone edge is at $kh = \pi h/a$, about 6 for $h/a = 2$. Most lattice modes are therefore invisible to the stress move, whatever the spacing.

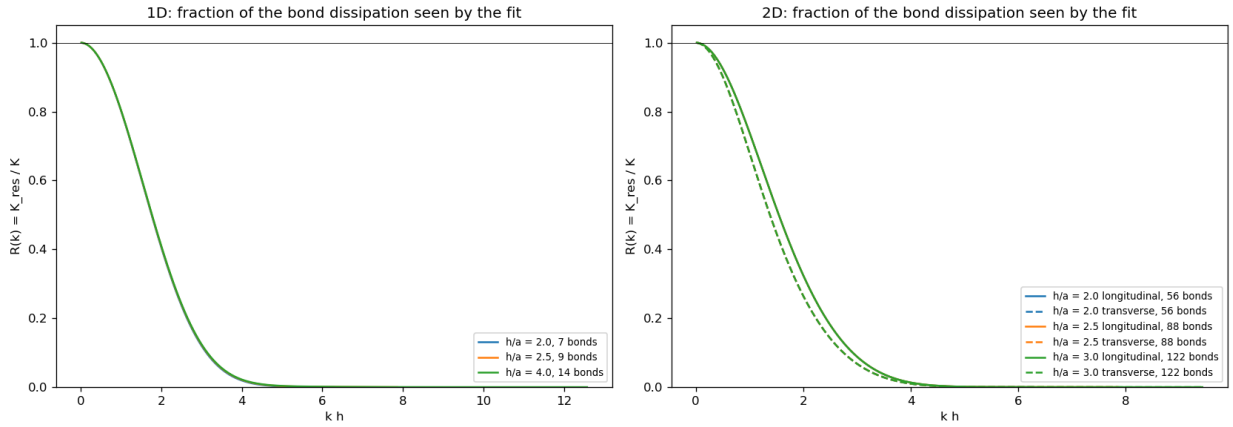


FIG. 1. The fraction $R(k) = K_{\rm res}(k)/K(k)$ of the bond dissipation seen by the fit, versus kh, for several h/a. Left: 1D. Right: 2D, longitudinal (solid) and transverse (dashed). The curves for different h/a coincide.

b. Frozen lattice. Fig. 2: the frozen lattice of `stochrep.tex` ($h/a = 2.5$, no Hamiltonian step), from zero momenta, with the stress move only. Each mode follows $1 - e^{-2\Gamma_{\rm res} t}$. The modes with $kh \lesssim 1$ thermalize as with both moves; those beyond $ka/\pi \simeq 0.5$ have barely moved by the end of the run. The kinetic energy reaches 0.22 of its equilibrium value, against 0.99 with both moves.

c. With the Hamiltonian step. On a frozen lattice the unresolved modes merely stay cold. With the Hamiltonian step on they are driven by the ideal SPH force, and nothing removes what it puts in. Fig. 3 and Fig. 4: $h/a = 2$, per-bond vote, plain map, to $t = 15$.

- *Stress and residual moves, $\lambda = 1$.* The spectrum is flat out to $kh \simeq 9$ and the kinetic energy settles at its equilibrium value. H drops once by about 0.3%, as thermal energy moves into particle-scale fluctuations that the smoothed H does not count. It is then constant until $t \simeq 8$, and afterwards creeps up by up to 0.1% by $t = 15$; $\sum_a e_a$ stays within 10^{-3}.

- *Stress move and deterministic residual, $\lambda = \infty$.* The lattice modes stay cold, the spectrum follows $T_{\rm eff}(k)$, and H and $\sum_a e_a$ are constant to about 10^{-4}.

- *Stress move only.* The modes with $kh \lesssim 3$ look right, but the lattice modes are heated by the Hamiltonian step and not damped. After $t \simeq 6$ the kinetic energy grows exponentially, by about a factor e per 3.5 time units, and reaches 4 to 7 times its equilibrium value at $t = 15$ in all four realizations; $\sum_a e_a$ grows with it, and the acceptance falls.

The same happens at $h/a = 2.5$ (one realization).

In all three variants the particles drift away from the lattice. Their rms displacement reaches about one lattice spacing by $t = 15$, mostly by collective motion. The local disorder reaches 0.18–0.27 a with $\lambda = 1$ or the stress move alone, and 0.11–0.15 a with $\lambda = \infty$.

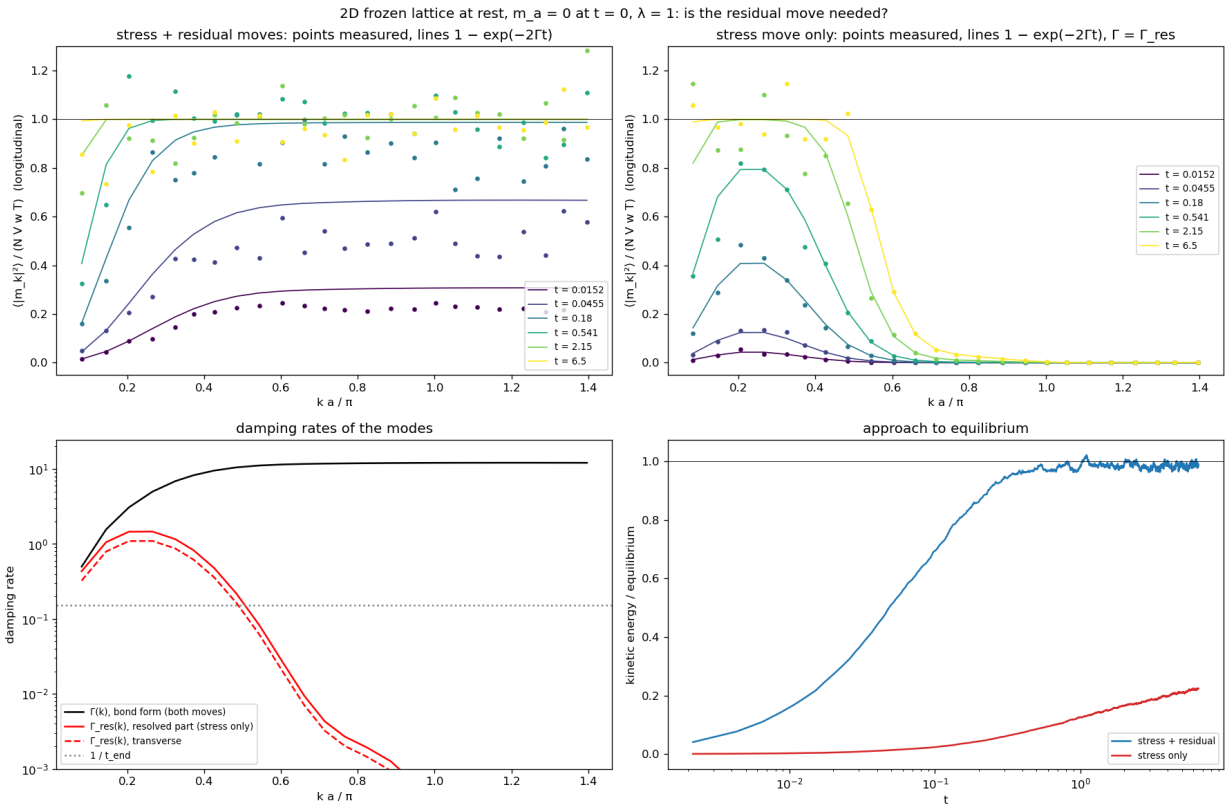


FIG. 2. 2D frozen lattice ($h/a = 2.5$) from zero momenta. Top: the spectrum at several times with both moves (left) and with the stress move only (right); points measured, lines $1 - e^{-2\Gamma t}$ with Γ and $\Gamma_{\rm res}$. Bottom left: the two rates. Bottom right: the kinetic energy.

d. Conclusion. The residual move is needed: it is what keeps the lattice modes under control once the particles move. For the modes with $kh \ll 1$ the stress move alone gives the right fluctuations, and the residual can be cold ($\lambda > 1$) or deterministic ($\lambda = \infty$) if the lattice modes need not fluctuate. The stress-only variant is dropped from here on.

III. LONGER RUNS: A SLOW HEATING

Fig. 5: the same setup to $t = 30$, $\lambda = 1$ and $\lambda = \infty$, four realizations each, per-bond vote, plain map.

- $\lambda = \infty$: the kinetic energy, H, $\sum_a e_a$ and the local disorder all level off; H and $\sum_a e_a$ change by less than 10^{-4}.

- $\lambda = 1$: the kinetic energy stays at its equilibrium value, but after $t \simeq 8$ both $\sum_a e_a$ and H rise, faster and faster, in all four runs: $\sum_a e_a$ by 0.24–0.45% and H by 0.3–0.5% from its minimum, by $t = 30$. The fluid heats slowly. The local disorder keeps growing, to 0.25–0.35 a.

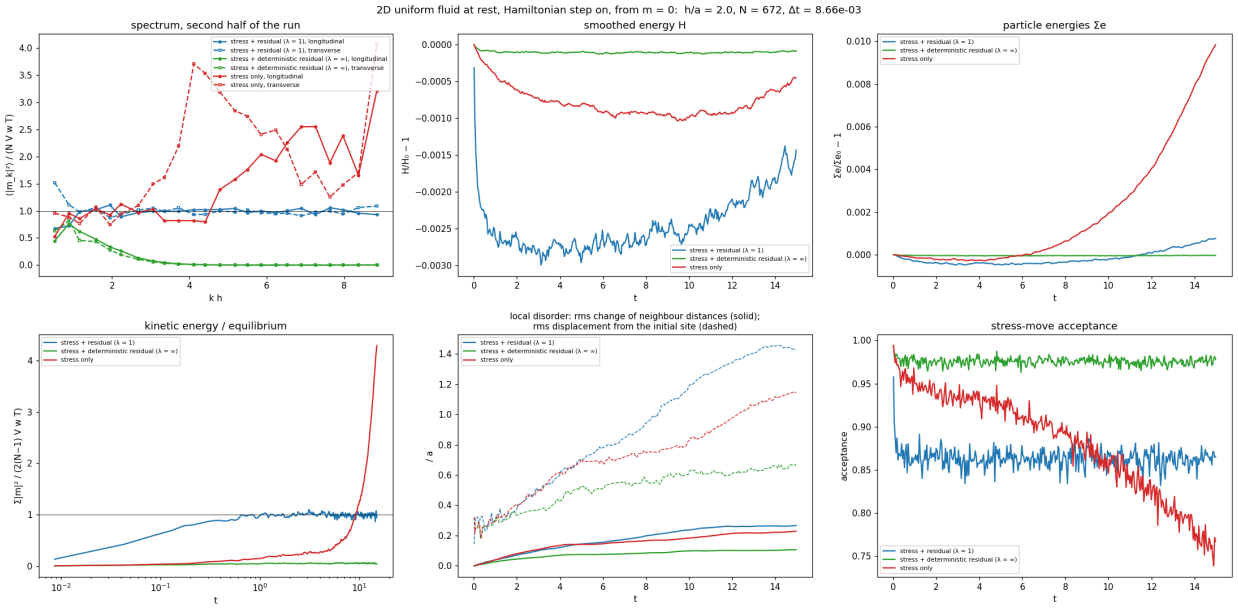


FIG. 3. $h/a = 2$, per-bond vote, plain map, one realization of each variant, to $t = 15$. Top: the spectrum over the second half of the run, the smoothed energy H, and the particle energies $\sum_a e_a$. Bottom: the kinetic energy; the local disorder (solid) and the rms displacement (dashed); the acceptance of the stress move.

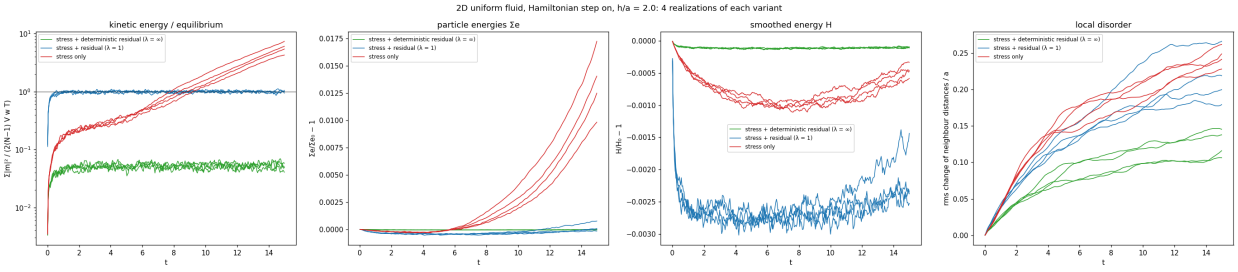


FIG. 4. As Fig. 3, four realizations of each variant: kinetic energy, particle energies, smoothed energy and local disorder.

IV. THE SINGLE VOTE

Fig. 6: as Sect. III with a single vote of the star in place of the per-bond vote. The residual move is accepted about 0.88 of the time instead of about 0.5, so the lattice modes are damped faster. The heating is about halved and starts later: at $t = 30$, $\sum_a e_a$ has changed by -0.05 to $+0.13\%$, and three of the four runs are close to flat. One run turns upward from $t \simeq 20$; it is also the run with the largest local disorder, $0.31\,a$. $\lambda = \infty$ is again flat.

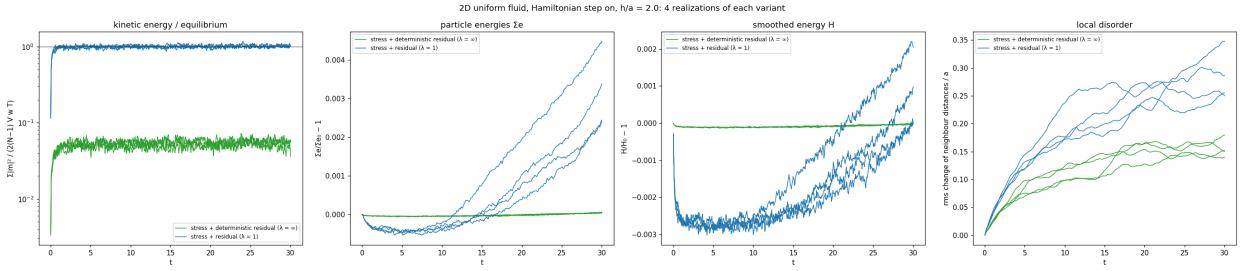


FIG. 5. $h/a = 2$, per-bond vote, plain map, to $t = 30$, four realizations of $\lambda = 1$ and $\lambda = \infty$: kinetic energy, particle energies, smoothed energy and local disorder.

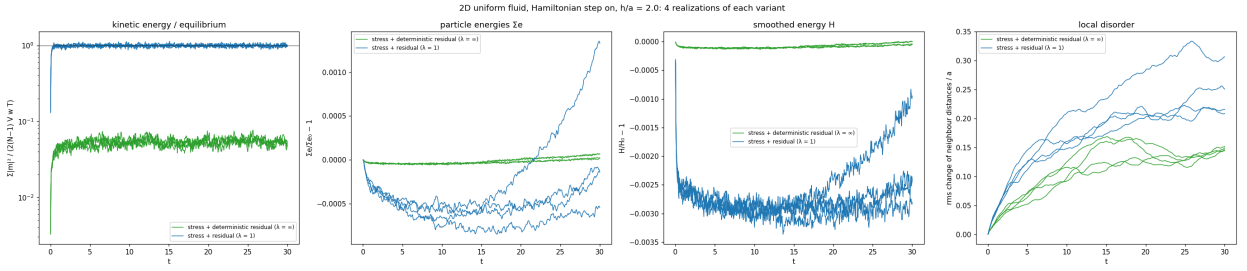


FIG. 6. As Fig. 5, with the single vote of the star.

V. WHERE THE ENERGY CHANGES

To locate the heating, each time step is split: the change of H and of $\sum_a e_a$ in the two stochastic half steps, and in the leapfrog, accumulated over the run. Fig. 7: $\lambda = 1$, single vote, at Δt and $\Delta t/2$, four realizations each.

- *The leapfrog conserves H to about* 10^{-7}, at both time steps. The integrator is not the source.
- *All the change of H comes from the stochastic half steps:* the initial drop, the plateau, and the late rise. The late rise is smaller at $\Delta t/2$, where all four runs stay within -2.5 to -2.9×10^{-3}.
- *The stochastic half steps lose $\sum_a e_a$ at a steady rate,* the same in every run: about -4×10^{-5} per unit time. Within a step each e_a is fixed at the frozen volume; afterwards s_a has changed by a random amount and the re-smoothing gives new volumes and energies. The first-order effect averages to zero; the second-order one does not, and accumulates linearly. It is an Itô-type term of the change of representation.
- *The leapfrog raises $\sum_a e_a$ slowly*, by 0.07–0.12% at $t = 30$. Since it conserves H, this is the gap $\sum_a e_a - H$ changing as the arrangement of the particles disorders.

VI. THE GRADIENT-CORRECTED CHANGE OF REPRESENTATION

The stochastic step changes representation twice per time step, between the smoothed variables of the leapfrog and the particle representation. The gradient-corrected map, which

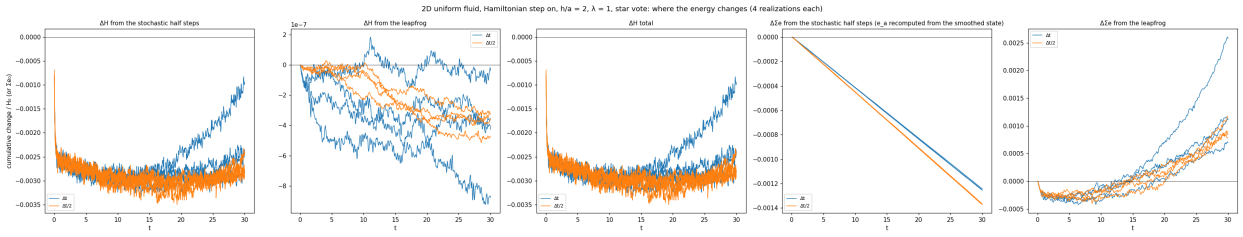


FIG. 7. $h/a = 2$, $\lambda = 1$, single vote, plain map, at Δt and $\Delta t/2$: cumulative changes of H from the stochastic half steps, from the leapfrog, and in total, and of $\sum_a e_a$ from the stochastic half steps and from the leapfrog.

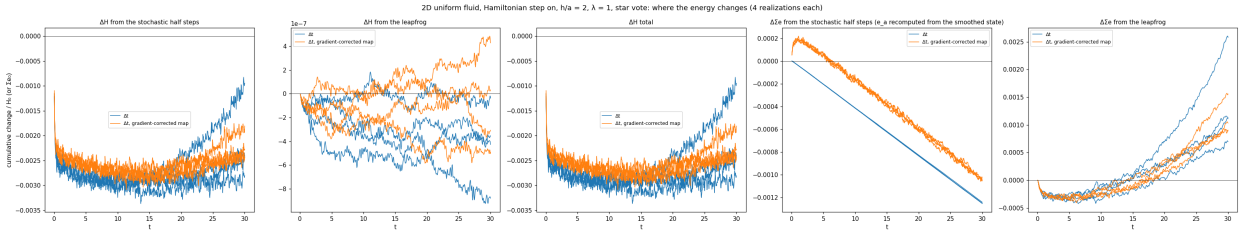


FIG. 8. As Fig. 7, plain map and gradient-corrected map at Δt.

removed most of the energy drift of the deterministic step (`changerep.tex`), is now used here too: the gradient energy g_a is frozen during the sweeps, so the sweeps and their equilibrium are unchanged, and the back map refits it. Fig. 8 and Fig. 9: $\lambda = 1$, single vote, Δt, four realizations.

- The kinetic energy stays at its equilibrium value.
- $\sum_a e_a$ stays within $\pm 5 \times 10^{-4}$ (plain map: -5×10^{-4} to $+1.3 \times 10^{-3}$).
- H, after the one-time drop of about 0.25%, stays within -2.9 to -1.8×10^{-3}, with a gentle late rise.
- The local disorder levels off at 0.18–0.28 a.
- The steady loss of $\sum_a e_a$ in the stochastic steps is unchanged: it comes from the random change of s_a, not from the gradient energy. The one-time drop of H is also unchanged: the correction captures the smooth part of the momentum field, not its thermal particle-scale fluctuations.

The remaining drift is small and smooth; a global thermostat can remove it if needed.

VII. LONG RUNS

Runs to $t = 100$, four realizations each, all at rest, to see how the long-time behaviour depends on the parameters. The reference is $h/a = 2$, $\lambda = 1$, single vote, gradient-corrected map, $\Delta t = 8.66 \times 10^{-3}$; each run changes one thing:

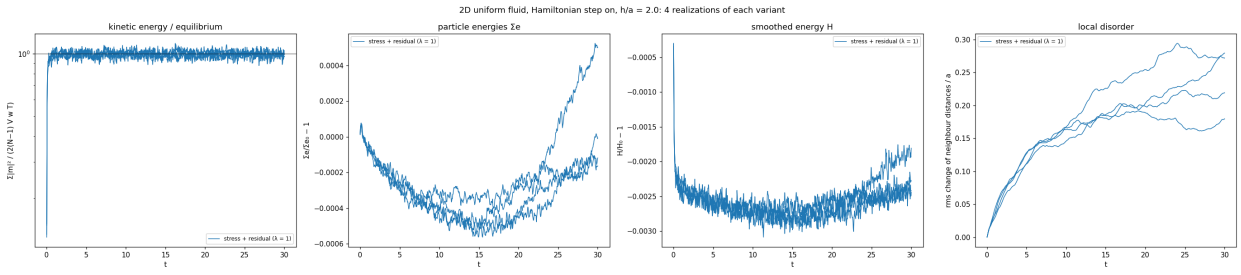


FIG. 9. $h/a = 2$, $\lambda = 1$, single vote, gradient-corrected map, to $t = 30$, four realizations: kinetic energy, particle energies, smoothed energy and local disorder.

reference	also $\lambda = \infty$
time step	$\Delta t/2$
vote	per bond
map	plain (no gradient correction)
viscosity	$\eta/s = 0.1$
fluctuations	$S = 10^5$: $s_a = 54$, thermal velocity 0.14
spacing	$h/a = 2.5$

Each set has four realizations; Fig. 10 shows them as bands, and Fig. 11 the set with larger fluctuations, which behaves differently.

set	kinetic energy	$\Delta \sum_a e_a / \sum_a e_a$	$\Delta H/H$	local disorder
reference	1.05	+1.2%	+0.9%	$0.27\,a$
$\lambda = \infty$	0.05	+0.09%	+0.09%	$0.19\,a$
$\Delta t/2$	1.03	+1.0%	+0.8%	$0.30\,a$
per-bond vote	1.10	+4.1%	+3.8%	$0.31\,a$
plain map	1.00	+1.45%	+1.2%	$0.30\,a$
$\eta/s = 0.1$	1.09	+4.1%	+3.9%	$0.31\,a$
$h/a = 2.5$	1.00	+0.7%	+0.3%	$0.51\,a$
$S = 10^5$	4×10^4	×50	×32	$3.4\,a$

(Values at $t = 100$, averaged over the realizations; the kinetic energy relative to its equilibrium value at the initial temperature.)

a. Slow heating everywhere. In every set with $\lambda = 1$ the energies rise, slowly at first and faster later. The local disorder levels off at 0.27–0.31 a for $h/a = 2$ after $t \simeq 30$, but the heating does not stop.

b. What it depends on.

- *Viscosity*: halving η/s makes the heating about 3.5 times faster.
- *Vote*: the per-bond vote, which rejects more residual moves and so damps the lattice modes more slowly, heats as fast as $\eta/s = 0.1$.
- *Time step*: halving Δt helps only a little (−15%).

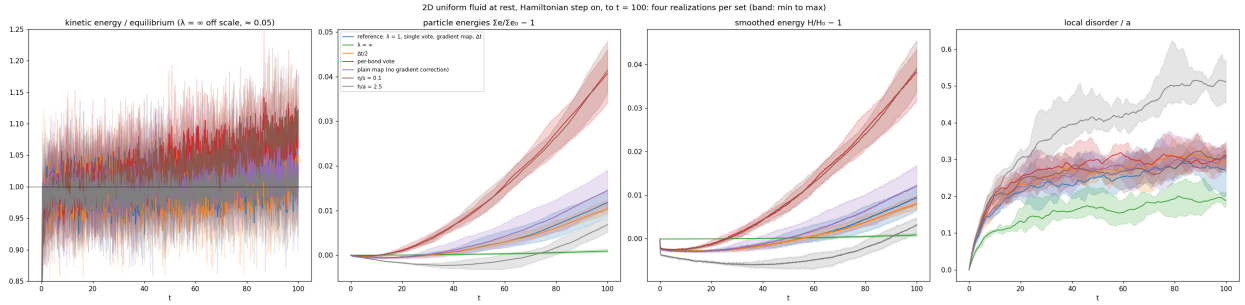


FIG. 10. Runs to $t = 100$, four realizations per set (band: minimum to maximum, line: mean). Kinetic energy, particle energies, smoothed energy and local disorder. The set with $S = 10^5$ is shown separately in Fig. 11.

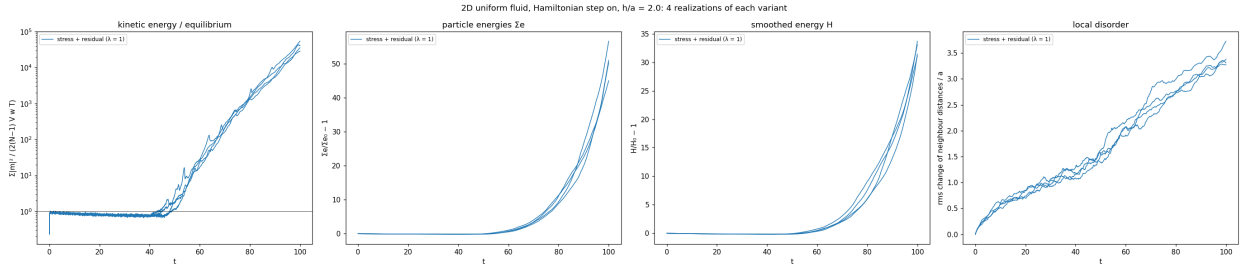


FIG. 11. $S = 10^5$ (entropy per particle 54, thermal velocity 0.14): four realizations. The lattice melts by $t \simeq 40$; the energies then grow exponentially.

- *Change of representation*: the gradient-corrected map heats about 20% less than the plain map.
- *Spacing*: $h/a = 2.5$ heats less, although its lattice disorders more.
- $\lambda = \infty$: the lattice modes are cold and the heating is ten times slower.
- *Fluctuations*: with $S = 10^5$ (entropy per particle 54, thermal velocity 0.14, three times larger) the lattice melts within $t \simeq 40$, the local disorder passing one lattice spacing, and then the energies grow exponentially, by a factor e in about 4.5 time units, to fifty times their initial value by $t = 100$. This is an instability, not a drift.

c. Interpretation (to be tested). The heating grows with the size of the thermal fluctuations and falls with the damping of the lattice modes (viscosity, acceptance, λ). This points to the lattice modes: the ideal step on a thermally agitated, disordering arrangement feeds them, and the viscous step removes it. When the fluctuations are too large compared with the damping, the lattice melts and the feeding wins. The energy split (Sect. V) places the change of H in the stochastic step; for the unstable set it is being measured. A criterion for stability, in terms of the thermal velocity, the viscosity and h, is the most important open question for production runs.

VIII. SUMMARY

- The residual move is needed once the particles move; without it the lattice modes run away.
- With both moves, $\lambda = 1$, the fluid thermalizes in about one time unit and stays at equilibrium, with a flat spectrum, out to $t = 30$.
- H drops once by about 0.3% as the particle-scale fluctuations fill in, then drifts slowly. The drift comes from the stochastic step, not from the leapfrog; it is reduced by the single vote, by a smaller time step and by the gradient-corrected map.
- With $\lambda = \infty$ the lattice modes stay cold and the energies are constant to 10^{-4} at $t = 30$ and to 10^{-3} at $t = 100$.
- To $t = 100$ every $\lambda = 1$ set heats slowly, faster for smaller viscosity or a vote that rejects more. With three times larger thermal velocities the lattice melts and the energies blow up: the scheme has a stability limit, set by the fluctuations against the damping, which must be found before production runs.